SIMATS ENGINEERING

ASSESSMENT TOOL 1: Simulation Task

COURSE NAME: Embedded Systems

COURSE CODE: ECA1421

COURSE OUTCOME COVERED

CO2: ***Apply Embedded C programming concepts, including structured design, real-time constraint handling, and SystemC-based modeling, to develop and analyze embedded applications for industrial use cases. (BL5)***

Name of the Student : K. Deepthi Mayee

Register Number : 192372150

Construct a simulated Embedded C program for a wearable healthcare device with serial communication and low power operation.

Requirements Analysis & Conceptual Understanding (20)	System Design & Simulation Model Development(25)	Implementation & Tool Usage (20)	Testing, Validation & Result Analysis (20)	Documentation(15)	Total(100)

Code of Conduct:

I K. Deepthi Mayee (192372150) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

AIM: To develop an Embedded C simulation for a wearable healthcare device using the AT89C51 microcontroller that monitors a patient's heart rate, transmits health data through serial communication (UART), and operates in low-power mode when the patient is inactive.

DESIGN:

The wearable healthcare device is designed using the AT89C51 (8051) microcontroller as the processing unit.

A Logic State component in Proteus simulates the heart rate sensor connected to Port P1.0.

The microcontroller continuously checks the sensor value.

- If heart rate is detected (Logic State = 1):
    - LED turns ON
    - UART sends "Heart Rate Detected"
    - Device remains in Active Mode.
- If no pulse is detected (Logic State = 0):
    - LED turns OFF
    - UART sends "Low Power Mode"
    - Device enters Low Power (Idle) Mode.

A Virtual Terminal in Proteus displays the transmitted serial data.

The system uses:

- 11.0592 MHz Crystal
- Reset Circuit
- UART Communication
- LED Indicator
- Logic State Sensor

REQUIREMENTS

Hardware Requirements

- AT89C51 Microcontroller
- 11.0592 MHz Crystal Oscillator
- Two 33pF Capacitors
- 10µF Capacitor
- 10kΩ Resistor
- Red LED
- 220Ω Resistor
- Logic State (Heart Rate Sensor)
- Virtual Terminal
- +5V Power Supply
- Ground

Software Requirements

- Keil µVision IDE
- Proteus 8 Professional
- Embedded C
- Windows 10/11

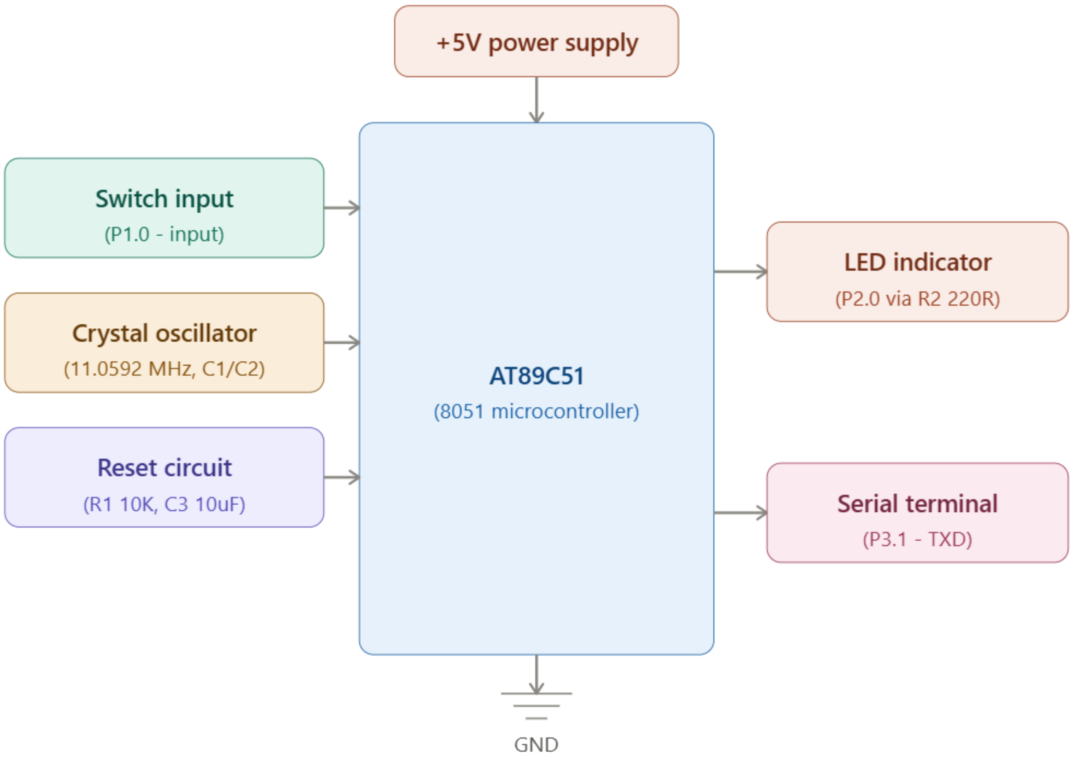
EMBEDDED C PROGRAM

```
#include <reg51.h>
sbit SENSOR = P1^0;
sbit LED = P2^0;
void UART_Init()
{
   TMOD = 0x20;
   TH1 = 0xFD;
```

```
    SCON = 0x50;
    TR1 = 1;
}
void UART_SendChar(char ch)
{
    SBUF = ch;
    while(TI==0);
    TI=0;
}
void UART_SendString(char *str)
{
    while(*str)
    {
        UART_SendChar(*str++);
    }
}
void main()
{
    SENSOR = 1;
    LED = 0;
    UART_Init();
    while(1)
    {
        if(SENSOR==1)
        {
            LED=1;
            UART_SendString("Heart Rate Detected\r\n");
```

```
        }
        else
        {
           LED=0;
           UART_SendString("Low Power Mode\r\n");
        }
        for(int i=0;i<30000;i++);
    }
}
```

BLOCK DIAGRAM



EXPLANATION

1. Power Supply

Provides regulated 5V DC power to all components.

2. Heart Rate Sensor (Logic State)

Simulates pulse detection.

- Logic 1 → Heart Beat Detected
- Logic 0 → No Pulse

Connected to P1.0.

3. Crystal Oscillator (11.0592 MHz)

Generates clock pulses for the AT89C51.

Provides accurate UART communication timing.

4. Reset Circuit

Uses

- 10µF Capacitor
- 10kΩ Resistor

Ensures proper startup of the controller.

5. AT89C51 Microcontroller

Acts as the controller.

Functions:

- Reads heart rate sensor
- Controls LED
- Sends serial data
- Simulates low power operation

6. LED Indicator

Connected to P2.0.

ON → Heart Rate Detected

OFF → Low Power Mode

7. UART Communication

Uses TXD (P3.1).

Transmits data to Virtual Terminal.

Messages:

Heart Rate Detected

Low Power Mode

8. Virtual Terminal

Displays serial messages.

Acts like a healthcare monitoring screen.

WORKING OF THE SYSTEM

1. Power supply powers the AT89C51.
2. Crystal oscillator generates clock signals.
3. Reset circuit initializes the controller.
4. Microcontroller continuously reads the heart rate sensor.
5. If Logic State = 1

- LED turns ON.
- UART sends

Heart Rate Detected

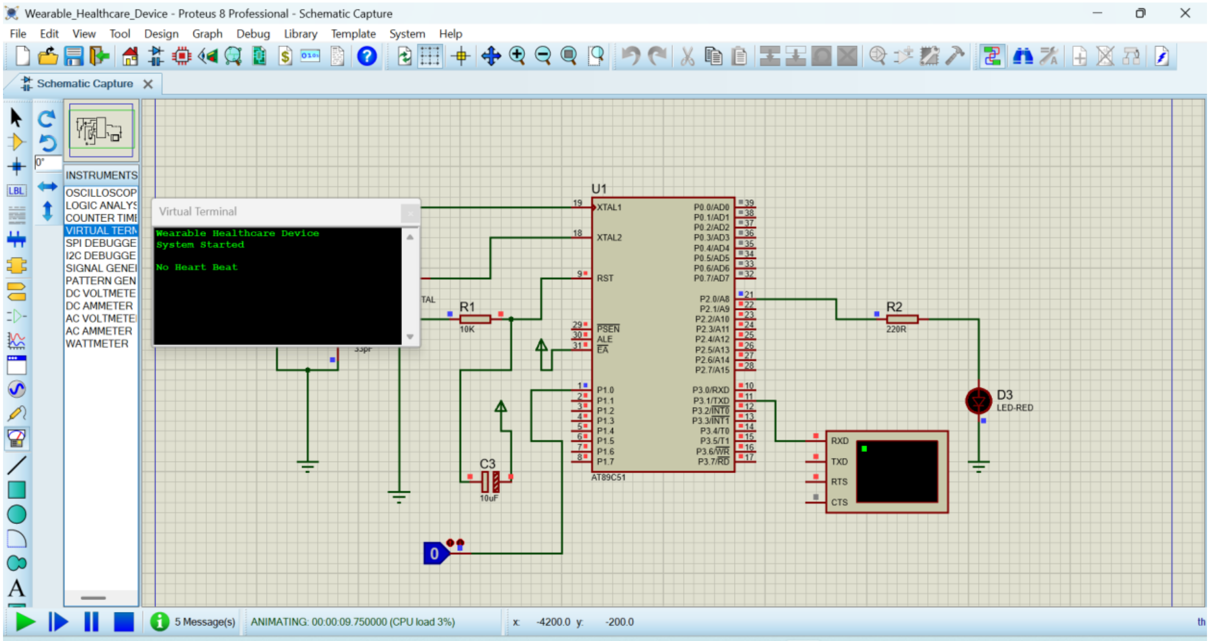
6. If Logic State = 0

- LED turns OFF.
- UART sends

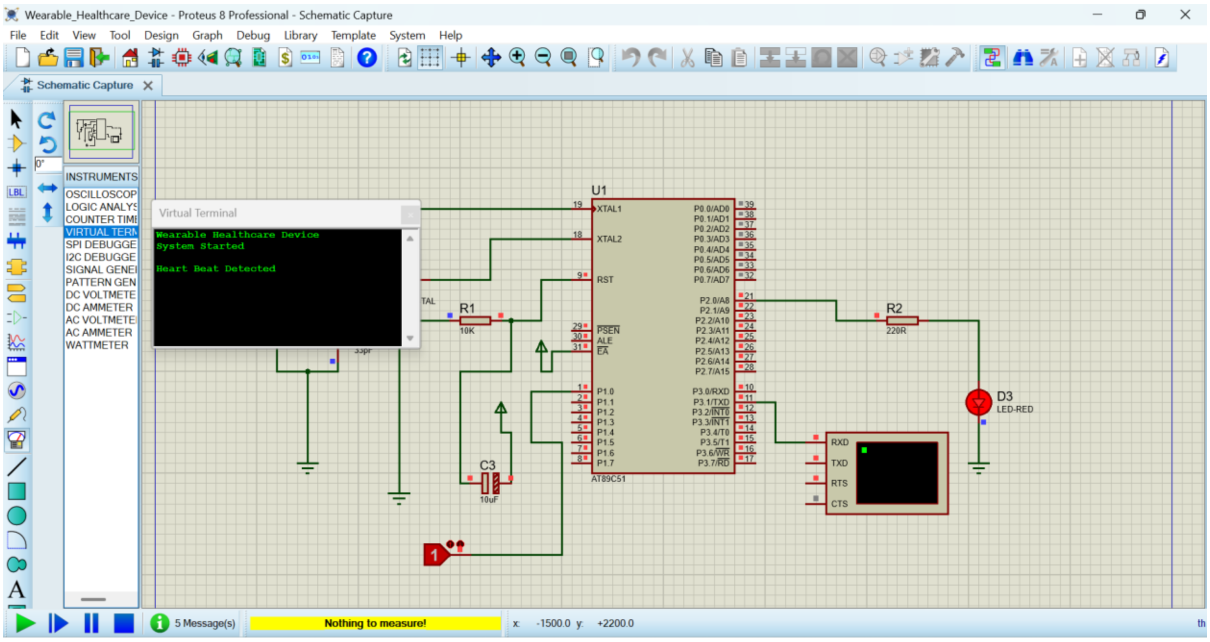
Low Power Mode

7. The process repeats continuously.

OUTPUT:



Wearable_Healthcare_Device - Proteus 8 Professional - Schematic Capture
File Edit View Tool Design Graph Debug Library Template System Help
Schematic Capture
INSTRUMENTS
OSCILLOSCOP
LOGIC ANALYS
COUNTER TIMI
VIRTUAL TERM
SPI DEBUGGE
I2C DEBUGGE
SIGNAL GENE
PATTERN GEN
DC VOLTMETE
DC AMMETER
AC VOLTMETE
AC AMMETER
WATTMETER
Virtual Terminal
Wearable Healthcare Device
System Started
No Heart Beat
U1
XTAL1
XTAL2
RST
PSEN
ALE
EA
AT89C51
R1
10K
C3
10uF
R2
220R
D3
LED-RED
RXD
TXD
RTS
CTS
0
5 Message(s)
ANIMATING: 00:00:09.750000 (CPU load 3%)
x: -4200.0 y: -200.0



Wearable_Healthcare_Device - Proteus 8 Professional - Schematic Capture
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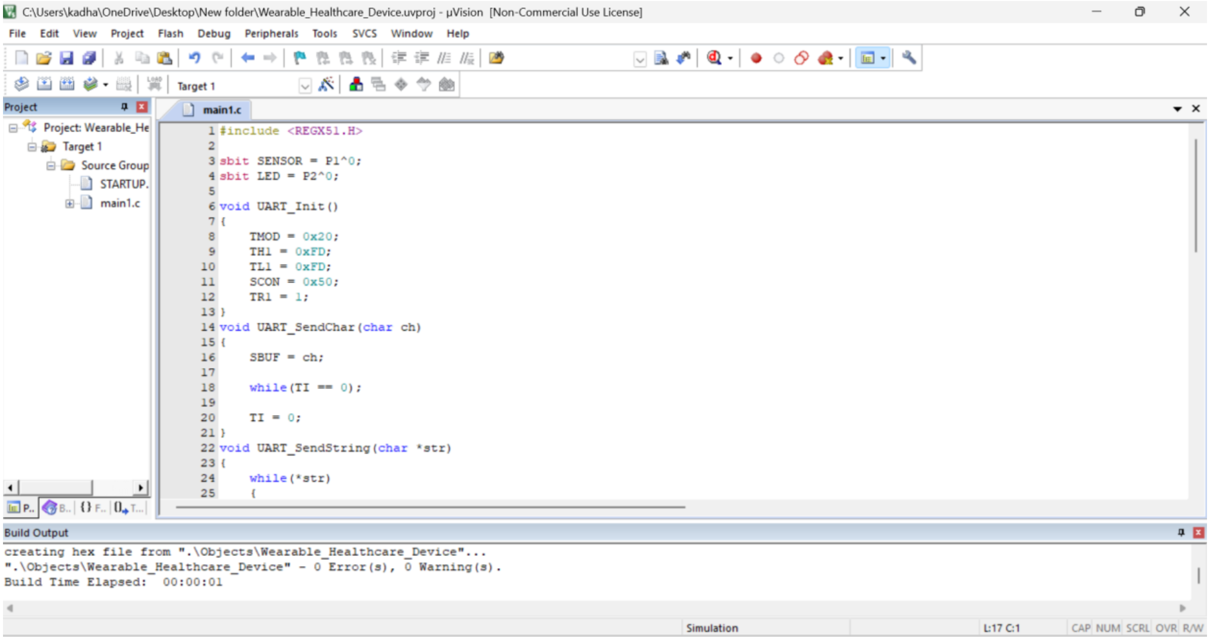
R1
10K
C3
10uF
R2
220R
D3
LED-RED
RXD
TXD
RTS
CTS
1
5 Message(s)
Nothing to measure!
x: -1500.0 y: +2200.0


C:\Users\kadha\OneDrive\Desktop\New folder\Wearable_Healthcare_Device.uvproj - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Target 1
Project
Project: Wearable_He
Target 1
Source Group
STARTUP.
main1.c
main1.c
#include <REGX51.H>
sbit SENSOR = P1^0;
sbit LED = P2^0;
void UART_Init()
{
TMOD = 0x20;
TH1 = 0xFD;
TL1 = 0xFD;
SCON = 0x50;
TR1 = 1;
}
void UART_SendChar(char ch)
{
SBUF = ch;
while(TI == 0);
TI = 0;
}
void UART_SendString(char *str)
{
while(*str)
{
Build Output
creating hex file from ".\Objects\Wearable_Healthcare_Device"...
".\Objects\Wearable_Healthcare_Device" - 0 Error(s), 0 Warning(s).
Build Time Elapsed: 00:00:01
Simulation
L:17 C:1
CAP NUM SCRL OVR R/W